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Artificial Intelligence in Ecuadorian SMEs: Drivers and Obstacles to Adoption

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Abstract: This study analyzes the current state of artificial intelligence (AI) adoption among micro-, small-, and medium-sized enterprises (MSMEs) in Ecuador, with a focus on its application across core business functions. Using a stratified random sample of 385 firms from the most representative economic sectors, a survey instrument was designed to assess three dimensions: access to AI-enabling conditions, degree of AI utilization, and organizational characteristics. The results reveal that AI adoption remains limited and highly concentrated in marketing-related functions, particularly in content generation and social media automation, with minimal implementation in finance, logistics, and human resource management. The study also identifies the main barriers hindering AI adoption. The lack of qualified professionals and the unavailability of structured databases emerged as the most critical obstacles, followed by limited financial capacity. One-way ANOVA and Kruskal–Wallis tests confirmed significant differences in AI adoption levels based on company size and sector, especially in areas such as inventory optimization and design prototyping. These findings highlight a gap between the potential of AI technologies and their real-world implementation in Ecuadorian MSMEs. They underscore the need for targeted strategies focused on workforce training, digital infrastructure development, and institutional support to promote broader and more effective AI integration.

Keywords: artificial intelligence; SMEs; Ecuador; business administration

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Academic Editors: Pejman Ebrahimi, Mihai Dascalu and Kostas Vergidis

Received: 22 March 2025
Revised: 12 May 2025
Accepted: 21 May 2025
Published: 27 May 2025

Citation: Pérez-Campdesuñer, R.; Sánchez-Rodríguez, A.; García-Vidal, G.; Martínez-Vivar, R.; De Miguel-Guzmán, M. Artificial Intelligence in Ecuadorian SMEs: Drivers and Obstacles to Adoption. *Information* **2025**, *16*, 443. https://doi.org/10.3390/info16060443

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1. Introduction

The development of machines designed to facilitate or replace human labor has been a persistent feature throughout the evolution of human civilization. In the realm of artificial intelligence (AI), early efforts were directed toward the creation of mechanical calculators, notably those proposed by Pascal in 1642 and Leibniz in 1672. Several decades later, Charles Babbage designed the Analytical Engine in the early 19th century, laying the groundwork for modern computing. This trajectory culminated in the construction of the first programmable computer, Colossus, in 1943.

In 1936, Alan Turing introduced the concept of the Turing Machine, establishing the theoretical foundation of modern computation. By 1950, he proposed the Turing Test as a method to evaluate whether a machine could emulate human intelligence. During this same decade, the first AI programs were developed to play chess and checkers, and in 1956,

John McCarthy coined the term "artificial intelligence". The following years witnessed significant milestones, including the creation of the first AI programming language in 1958 and the development of ELIZA in 1966, one of the earliest chatbots capable of simulating a basic human conversation.

In the following decades, the momentum of AI development slowed due to limited computational capacity and data availability. However, the final two decades of the 20th century witnessed significant progress in mathematical foundations, particularly through the advancement of neural networks. A key milestone was the introduction of the backpropagation algorithm in 1986 [1], which became essential for training neural networks, an underpinning technique in today's AI systems. In the early 21st century, substantial improvements were achieved in both software development and access to large-scale datasets. For instance, in 2012, deep neural networks revolutionized image recognition, and by 2014, the emergence of Generative Adversarial Networks (GANs) [2] enabled the creation of highly realistic synthetic images.

As the third decade of the 21st century began, AI adoption entered an exponential growth phase. Advanced models capable of understanding and generating natural language emerged, alongside an increasing proliferation of AI systems and applications across diverse domains. Although precise figures are difficult to determine due to the rapid pace of development, estimates suggest there are currently between 50 and 100 active large language models (LLMs) [3] developed by companies such as OpenAI, Google DeepMind, Meta, Anthropic, Alibaba, and Nvidia. For the generation of images, audio, and video [4], over 200 commercial models are available. Regarding virtual assistants and chatbots [5], several thousand are currently in operation. Platforms such as Notion AI, Grammarly, Jasper AI, Copy.ai, GitHub Copilot, and Claude AI offer a wide range of AI-powered functionalities. In total, it is estimated that there are between 10,000 and 100,000 AI tools and models available online, encompassing commercial platforms, chatbots, automation systems, open-source models, and other intelligent applications.

Business administration represents a fertile field for the application of artificial intelligence (AI) [6]. Since the development of the Mathematical School within administrative sciences, managerial practices have become a promising area for AI integration. The application of AI in business administration is diverse and can be categorized into the following areas:

- Data-driven decision-making: AI enables rational decision-making supported by data, models, and projections. It facilitates the analysis of large data volumes (Big Data), the detection of patterns, trends, and anomalies, the generation of predictions (e.g., sales, employee turnover, demand), and the automation of repetitive or complex decisions.
- Resource and process optimization: One of the classical objectives of management is to optimize resource utilization. AI contributes through optimization algorithms for logistics, inventory, and task allocation, as well as predictive models that enhance operational planning. Additionally, Decision Support Systems (DSS) powered by AI offer solutions for complex scenarios.
- Automation and organizational efficiency: Business administration seeks to improve organizational efficiency. AI enables the automation of administrative processes through Robotic Process Automation (RPA), reduces customer response times via chatbots and virtual assistants, and enhances human resource management through performance analysis and intelligent recruitment.
- Strategic planning and innovation: AI assists managers in simulating economic and organizational scenarios, making more agile and data-driven decisions, and generating competitive advantage through innovation in business models.

However, it is unlikely that AI adoption will follow a uniform pattern, as it depends on multiple variables such as company size, economic performance and purchasing power, prevailing managerial mindset, digital skills, and organizational culture, among others.

Specifically, the context of micro-, small-, and medium-sized enterprises (MSMEs) presents significant challenges in integrating AI, as their managers do not always possess a high level of professional administrative competencies. Additionally, the short lifespan of these organizations and the high turnover of their workforce often hinder the development of a strong organizational culture. Furthermore, MSMEs typically operate with substantial financial constraints. These characteristics have been analyzed by the authors in previous studies [7].

Globally, the economic landscape is largely composed of MSMEs, which serve as the primary source of income and employment for a significant share of the economically active population. This trend is especially pronounced in developing countries [8]. In Ecuador, for instance, MSMEs contribute over 25% of the national GDP and generate employment for approximately 70% of the economically active population. Nationwide, there are 6,089,518 registered businesses, of which micro and small enterprises account for 90.4%. Their sectoral distribution is as follows: agriculture (98.5%), extractive industries (94.4%), manufacturing (97.4%), construction (97.1%), commerce (97.8%), and services (98.3%) [9].

Moreover, in Ecuador, it has been established that SMEs exhibit a high mortality rate. Most entrepreneurs within these organizations do not have formal training as professional administrators and face significant financial limitations.

This study aims to address the following research questions: What is the degree of adoption of AI technologies across the various administrative subsystems or functions of small- and medium-sized enterprises (SMEs) in Ecuador? And what are the main barriers that hinder their integration within these organizations?

To answer these research questions, this study proposes an analysis of the relevant literature, followed by the development of the research methodology, which outlines the variables to be measured, the general characteristics of the designed instruments, the characterization of the population and the sample, as well as the methods used for data processing. The findings are then presented, along with the main conclusions derived from the analysis. Additionally, the study identifies research limitations and suggests potential future research lines. Finally, the references supporting this study are listed to provide a comprehensive theoretical foundation.

2. Literature Review

Although the foundational principles of administrative sciences can be traced back to pre-20th century practices—such as those rooted in military strategy or the hierarchical structures of the Catholic Church—the formal recognition of business administration as an independent scientific discipline emerged in the 20th century. Jurado [10] offers a comprehensive synthesis of the major schools of management thought. Beyond the foundational contributions of Frederick Taylor and the subsequent developments by Henri Fayol, significant advancements emerged from the Human Relations School, which emphasized the critical role of human resources within organizations. The Mathematical School also made notable contributions by introducing quantitative methods to optimize production processes and enhance decision-making.

The Systems School [11] brought a transformative perspective by conceptualizing the organization as an open system composed of interconnected subsystems, processes, and functions. These components interact with one another and are directly influenced by the external environment, thereby shaping organizational performance.

Similarly, adopting an ecosystemic perspective, contingency theory [12] posits that organizations must implement administrative systems that are shaped by both internal and external environmental factors. The theoretical contributions of the previously mentioned schools have been further enriched by complexity theory [13], which explores how organizations are influenced by dynamic processes of change.

Concurrently with the evolution of these theoretical schools, the core functional areas of business administration began to develop their own methodological frameworks as adaptive mechanisms in response to internal transformations and external environmental shifts. For instance, organizations have transitioned from 100% quality inspection toward comprehensive quality management systems [14]; from traditional sales management to relational and customer-centric marketing [15]; from process management to integrated, extended supply chain management [16]; and from basic personnel administration to strategic human resource and talent management [17].

Financial and economic management [18] can be approached as an integrated system that constitutes a fundamental pillar for most organizations. Alternatively, it may be understood from a more fragmented perspective, composed of multiple subsystems such as accounting, economics, finance, and even internal control or auditing functions.

Within this framework, it is important to note that not all administrative subsystems have evolved uniformly or over the same temporal horizon. Some did not originate as independent functions but rather emerged through the transformation or expansion of existing subsystems. Examples include innovation management [19], intellectual property management, information or technology management, and environmental management. The latter [20] arose in response to the growing need to preserve resources and protect the natural environment.

In practice, quality management may be structured as an independent area or integrated within other functional domains, particularly logistics or marketing. In some cases, it evolves into a broader framework, total quality management [21], which is conceived as a holistic outcome that reflects the performance of the organization as a whole.

A search in the Scopus database using the keyword artificial intelligence yields a total of 644,203 publications. The earliest record dates back to 1960, authored by Shubik [22], highlighting the potential applications of AI in games and simulation processes. Since then, the number of publications on the subject has shown a steady and significant upward trend.

Figure 1 presents the evolution of AI-related publications from their inception through the year 2024. In the upper section of the curve, it can be observed that publication trends remained relatively linear until 2022. However, it is important to note that from 1960 to that year, at least one publication was recorded annually, exhibiting an overall upward trend. Between 2002 and 2020, a noticeable acceleration in the number of publications occurred, and from 2020 onward, the growth rate has increased significantly.

To better illustrate this dynamic, the lower part of Figure 1 decomposes the trend into three distinct phases. A linear equation offers the best fit for the first stage, whereas exponential models more accurately capture the behavior of the latter two phases. In both of these stages, the exponential growth coefficients are high, further reinforcing the notion of rapid and accelerating expansion in AI-related research.

It is important to clarify that not all software applications can be classified as manifestations of artificial intelligence. To determine whether a given application qualifies as AI, it must meet certain essential characteristics [23]:

- It improves its performance over time by leveraging machine learning algorithms to learn from data and optimize outcomes.
- It adapts its behavior in response to new conditions or data without requiring manual reprogramming.

- It analyzes information and generates responses without direct human intervention.
- It is capable of interpreting text, voice, or images in a manner similar to human perception.
- It performs tasks that previously required human intelligence, such as text analysis, decision-making, or pattern recognition.

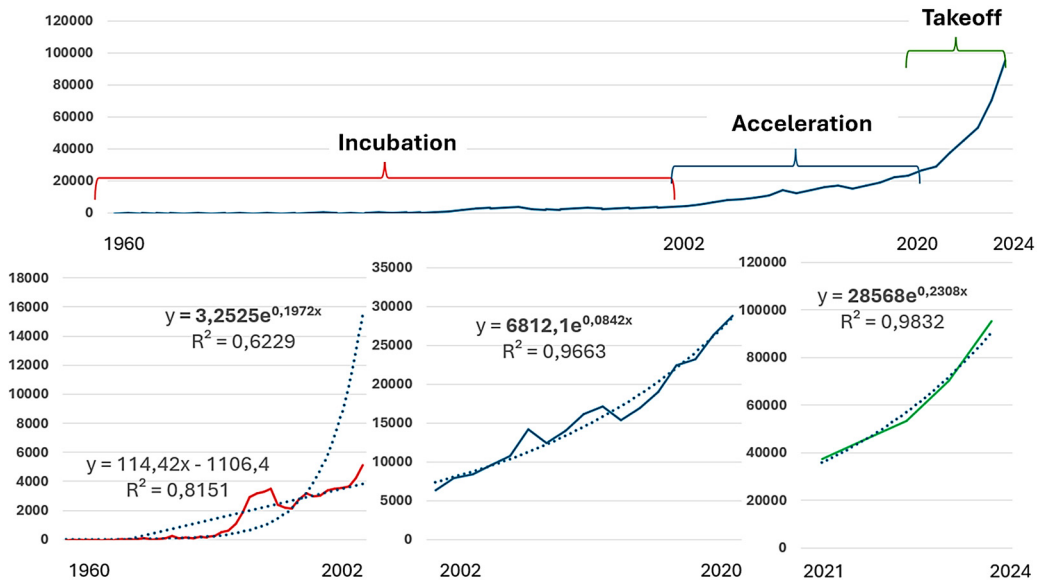

Figure 1. Annual publication trends on artificial intelligence.

The forms, operational methods, and applications of artificial intelligence in business environments are diverse. Below are some of the most widely recognized approaches.

Neural networks are computational models inspired by the functioning of the human brain. They are used in artificial intelligence (AI) and machine learning (ML) to recognize patterns, support decision-making, and learn from data. Neural networks have evolved through several development stages since their inception in the 1940s, driven by advances in technology, data availability, and algorithmic techniques. They are widely applicable in administrative sciences, contributing to decision-making [24], process optimization [25], and data analysis [26].

Genetic algorithms (GAs) represent another branch of AI. Inspired by biological evolution, these algorithms are used to solve complex optimization and search problems. In business administration, they are applied to optimize management variables [27].

Large language models (LLMs) are AI systems designed to understand, process, and generate natural language [28]. Trained on massive datasets and based on advanced neural network architectures, LLMs have multiple applications in management, including the development of virtual assistants [29], automated writing [30], data analysis [31], machine translation [32], and customer support [33]. All these uses contribute to increased efficiency and effectiveness in business processes and managerial decision-making.

Social media automation using artificial intelligence (AI) [34] is employed to enhance efficiency in content management, user interaction, and data analysis across platforms such as Facebook, Instagram, Twitter (X), LinkedIn, and TikTok.

AI-powered code assistants [35] are used to optimize, automate, and enhance developer productivity in tasks such as code writing, debugging, and documentation. These tools are particularly valuable for SMEs and technology startups seeking to accelerate development processes without incurring additional personnel costs.

The integration of AI into administrative subsystems can be achieved through various functions, depending on their objectives and the potential of AI to meet these goals. In

general, all business functions or administrative subsystems must systematically address the process of data-driven decision-making. However, this process materializes differently depending on the characteristics of each business function. For example:

- Logistics and process management: This function must ensure the optimization of resources throughout the processes of procurement, storage, production, and distribution [36]. In this context, AI contributes through optimization algorithms for logistics, inventory, and task allocation, as well as predictive models that improve operational planning. Additionally, Decision Support Systems (DSS) powered by AI provide solutions for complex scenarios.
- Production process management: This area involves specific actions such as product and process design [37], quality control [38], and process automation [39], where innovation plays a leading role. Moreover, the goal is to automate production to enhance productivity and control.
- Marketing and commercialization function: This area faces various challenges as part of its performance, including market research and customer analysis, monitoring the market behavior of products, competitors, and other decision variables [40]. Additionally, it involves designing and implementing advertising strategies and evaluating their impact [41].
- Financial and economic management must ensure the existence of an optimal financial flow, which requires the execution of multiple operations, including investment evaluation, expense forecasting and control, and auditing processes to identify potential non-compliance with established standards [42].
- Human resource management must ensure the timely availability of the necessary workforce with the required competencies. To achieve this, it must carry out various operations, such as human resource planning, recruitment and selection, personnel training, performance evaluation, and payroll management, among others [43].
- Security management is one of the key functions of business management, recognized since the early contributions of Henry Fayol. In this context, various activities are developed, including data protection, monitoring of suspicious activities both inside and outside the organization, among other security measures [44].
- The implementation of AI in these areas can generate significant benefits, including cost reduction, increased productivity, more efficient decision-making, and personalized customer service, ultimately leading to higher levels of business competitiveness [45].

SMEs have characteristics that distinguish them from other types of companies, not only because they employ fewer workers but also because their business subsystems must operate more dynamically and with narrower tolerance margins. Additionally, they face greater capacity and financial constraints, making it more challenging to establish a solid organizational culture.

A search in Scopus using the query TITLE-ABS-KEY (sme) AND TITLE-ABS-KEY (ai) identified only 326 publications, which are predominantly concentrated in Asia [46], Europe [47], and the United States of America. Figure 2 illustrates this distribution, highlighting that research collaboration networks on this topic are mainly situated in these regions. Nevertheless, it is noteworthy that, albeit to a lesser extent, some publications have also been reported from Africa [48] and the Americas [49].

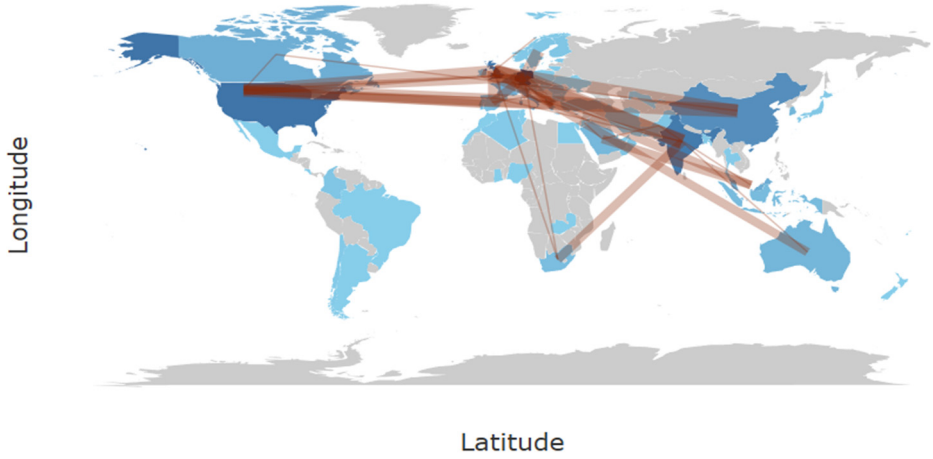

Figure 2. Geographical distribution of research on SMEs and AI.

Several of these publications assess the level of AI introduction or adoption in SMEs [50] and the impact generated by these technologies [51], as well as the barriers faced by organizations [52] and the influence of the existing legal framework [52] during the implementation of these technologies in this type of organization.

Research conducted on SMEs aligns with the administrative science functions previously discussed. For instance, studies have evaluated AI utilization in financial management [53], accounting [54], internal control [55], marketing [56], supply chain management, and general logistics [57], and specifically in the production process [58] and innovation process [59]. Additionally, there has been in-depth exploration of human talent management [60], covering various aspects such as training, selection processes, and occupational safety and health. Specific studies have also assessed the use of AI in software application development [61] and cybersecurity measures [62].

Table 1 presents a summary of how various AI applications have been integrated into different administrative science functions based on previous research aimed at specific functional improvements.

Table 1. Potential AI applications by business administration subsystems.

Business Function	Application Area	Applicable AI Tools	Reference Authors
Marketing	Customer service and sales	ChatGPT, Zendesk AI, Freshdesk AI, Drift, Tidio	Rawindaran et al. [63]
	Social media automation	ManyChat, Sprinklr AI, Hootsuite AI, Buffer	Dangi et al. [64]
	Customer data analytics	HubSpot AI, Salesforce Einstein, Zoho CRM AI	Santana and Cánepa [65]
	Content generation	Jasper AI, Copy.ai, Canva AI, Writesonic, Runway AI	Thapa et al. [66]
	Advertising optimization	Google Ads AI, Meta (Facebook) Ads AI, Adzooma	Haji and Stock [67]
	Market analysis	Crayon, Brandwatch AI, Nielsen AI	Wang [68]
	Trend analysis	TrendAI, Crayon, SEMrush AI	Wang et al. [69]
Financial Management	Accounting automation	QuickBooks AI, Xero, Zoho Books, Sage Intacct AI	Artene and Domil [70]
	Financial forecasting	Fathom, Domo AI, Syft Analytics	Fu et al. [71]
	Fraud detection	Stripe Radar, Darktrace, Kount AI	Hafez et al. [72]

Table 1. *Cont.*

Business Function	Application Area	Applicable AI Tools	Reference Authors
Human Resource Management	Recruitment and candidate selection	LinkedIn Recruiter AI, BambooHR AI, Workday AI	Trovão et al. [73]
	Payroll management	Deel, Gusto, Rippling AI	Susitra et al. [74]
	Employee performance evaluation	Lattice, 15Five AI, Peakon	Ravisankar et al. [75]
Logistics Management	Inventory optimization	TradeGecko AI, NetSuite AI, Inventory Planner	Tsintotas et al. [76]
	Order and shipping automation	ShipBob AI, Route4Me AI, Flexport AI	Patnaik and Patnaik [77]
	Route optimization	Onfleet AI, Routific, Google Maps AI	Chen and Zhu [78]
Innovation	Design and prototyping	Figma AI, Adobe Firefly, Runway AI	Tyagi et al. [79]
Production	Code assistants	GitHub Copilot, Tabnine, Replit Ghostwriter	Li et al. [80]
Security and Internal Control	Cybersecurity	Darktrace, CrowdStrike, Microsoft Defender AI	Zhang et al. [81]
	Data protection	IBM Security AI, Tessian AI, Symantec AI	Almotiri [82]
	Suspicious activity monitoring	Vectra AI, Cisco Secure AI, Splunk AI	Raja et al. [83]

Research on the development of artificial intelligence (AI) in the context of small and medium-sized enterprises (SMEs) is diverse and addresses a variety of perspectives. Some studies [84] offer a general assessment of the background and consequences of introducing AI technologies in these organizations, while others [85] focus on characterizing the current state of the art regarding AI adoption in SMEs at the international level.

Similarly, certain studies [86] explore the underlying motivations that drive SMEs to adopt AI, whereas others [87] examine the dynamics of technology implementation within the target organizations. Additional research [88] evaluates potential strategies or alternatives that may facilitate the effective integration of AI technologies.

On another front, some studies are dedicated to analyzing specific variables that influence technology adoption. For instance, they assess how AI can enhance organizational resilience [89], investigate its impact on knowledge management [90], explore the role of the external environment in shaping adoption decisions and its influence on innovation capacity [91], and analyze its application potential in marketing functions [92].

Similar lines of inquiry have been developed with a focus on specific business areas such as supply chain management [93], product design [94], human resource management [95], and occupational health and safety [96], to name a few.

However, it is important to highlight that most of these studies are concentrated in developed countries, where the conditions for AI adoption are more favorable, and fewer barriers exist in terms of access to financing, technological infrastructure, and skilled human capital. For this reason, the majority of research comes from countries such as China [97], India [98], and the United States [99], among others [100]. In contrast, the integration of these technologies in SMEs from developing nations progresses at a slower pace due to the presence of multiple structural obstacles.

3. Materials and Methods

The research was conducted following the steps outlined below:

3.1. Design of the Research Instrument

A research instrument was developed to assess three key dimensions: (a) the ease of access to AI tools for administrative functions, (b) the degree of AI utilization across various business areas, and (c) the characterization of companies according to their size and the sector in which they operate.

To assess access conditions, the following variables were evaluated:

- Availability of computing resources to access AI tools;
- Availability of internet connectivity for AI access;
- Availability of databases suitable for AI use;
- Availability of professionals with the knowledge to operate AI systems;
- Financial capacity to implement AI solutions.

To evaluate the level of AI utilization, organizations were asked whether they employed any of the AI applications listed in Table 2, which are associated with general business administration functions outlined in Table 1.

Table 2. Potential AI applications across business administration functions.

Business Function	AI Application Areas	Business Function	AI Application Areas
Marketing	Customer Service and Sales Social media automation Customer data analysis Content generation	Internal Control	Accounting automation Financial forecasting Fraud detection Accounting automation
	Advertising optimization Market analysis Trend analysis	Human Resource Management	Candidate selection Payroll management Employee performance evaluation
Logistics	Inventory optimization Order and shipping automation Route optimization Design and prototyping	Information Technology	Code assistants Cybersecurity Data protection Suspicious activity monitoring

In addition, for each AI function listed in the previous table, respondents were asked to specify which particular AI tool or system was being used. At the end of the questionnaire, organizations were asked to indicate their type: micro-, small-, or medium-sized enterprise.

All questions were structured using a dichotomous (yes/no) scale, except for the third question, which used a multiple-choice classification format.

The designed instrument was evaluated by seven experts, all of whom are university professors with over ten years of experience in research and teaching in the fields of computer science and administrative sciences. The experts assessed the instrument's face and content validity. Overall, they agreed that the instrument did not present any limitations affecting its face validity, as it was clearly structured and easy to understand. Similarly, there was consensus that the variables included were appropriate and aligned with the research objectives.

3.2. Population and Sample Characterization

Table 3 presents a general characterization of the population. Medium-sized enterprises classified as Type B, defined as those with 100 to 200 employees, were excluded from the study, as they were considered too large to be compared meaningfully with the rest of the enterprises in this category.

Table 3. General characterization of population and sample size.

Sector	Micro	Small	Medium	Total	% Share	Cumulative %	Sample Size
Services	486,715	19,320	2415	508,450	39.64	39.64	177
Commerce	364,638	17,477	3618	385,733	30.08	69.72	134
Information Tech	125,786	12,864	2204	140,854	10.98	80.70	49
Industry	67,653	3231	646	71,530	5.58	86.28	25
Education	28,124	5169	936	34,229	2.67	88.95	0
Agriculture	28,764	3793	830	33,387	2.60	91.55	0
Construction	27,764	3654	475	31,893	2.49	94.04	0
Health	27,172	1744	235	29,151	2.27	96.31	0
Textile	26,061	655	94	26,810	2.09	98.40	0
Arts	14,821	734	78	15,633	1.22	99.62	0
Finance	2014	806	145	2965	0.23	99.85	0
Mining	1273	311	72	1656	0.13	99.98	0
Communication	231	4	2	237	0.02	100.00	0
Total	1,201,016	69,762	11,750	1,282,528			385

For sample selection, the Pareto principle was initially applied, focusing exclusively on the top four sectors, which together represent over 86% of the total MSME population. A stratified random sampling method was then employed, based on an estimated population size of 1,106,567 enterprises. Given that this figure exceeds 100,000, the population was considered infinite for sampling purposes, and Equation (1) was applied accordingly.

$$n = \frac{p * q * k^2}{e^2} = \frac{0.5 * 0.5 * 1.96^2}{0.05^2} = 385 \quad (1)$$

where:

n: Sample size;

k: Critical value from the standard normal distribution for the desired confidence level (e.g., 1.96 for 95% confidence, 2.58 for 99%);

p: Expected proportion of success (if unknown, 0.5 is used to maximize variability);

q = 1 − p: Proportion of failure;

e: Acceptable margin of error (in decimal form, e.g., 0.05 for 5%).

Based on the total sample size, the corresponding sample size for each of the four sectors analyzed was determined proportionally, as shown in Table 2.

3.3. Data Processing

The collected data were processed using SPSS version 25. The analysis began with descriptive statistics, including absolute and cumulative frequencies. In addition to the general characterization of the variable's behavior, an indicator of AI adoption intensity was determined. The intensity of adoption was not measured by the frequency of AI use, but rather by the diversity of AI applications across different functions or the variety of uses within a single function. For this purpose, Equation (2) was applied.

$$\text{IIIA} = \frac{\sum_{I=1}^{21} \text{Types of AI Utilization}}{21 * n} \quad (2)$$

where:

IIIA: Intensity of artificial intelligence adoption;

n: Sample size.

Subsequently, a one-way ANOVA test was conducted to determine whether significant differences existed in variable behavior based on company sector or size. Additionally, a

Kruskal–Wallis test was conducted to assess the independence of the samples. Finally, a logistic regression analysis was performed, with AI adoption as the dependent variable, coded dichotomously: a value of 1 was assigned if the company used at least one AI tool, and 0 otherwise. The barriers to adoption were treated as independent variables. To verify the null hypothesis—"the model fits the data well (no significant differences between observed and expected values)"—the following metrics were employed: −2 Log Likelihood, Cox and Snell R Square, and Nagelkerke R Square, along with the significance level of the Hosmer–Lemeshow test ($p > 0.005$).

4. Results

The initial phase of the analysis revealed the presence of several AI applications across business functions. As shown in Figure 3, there are certain functional areas—particularly in IT-related processes—where AI adoption is virtually nonexistent. These include human resource management and internal control.

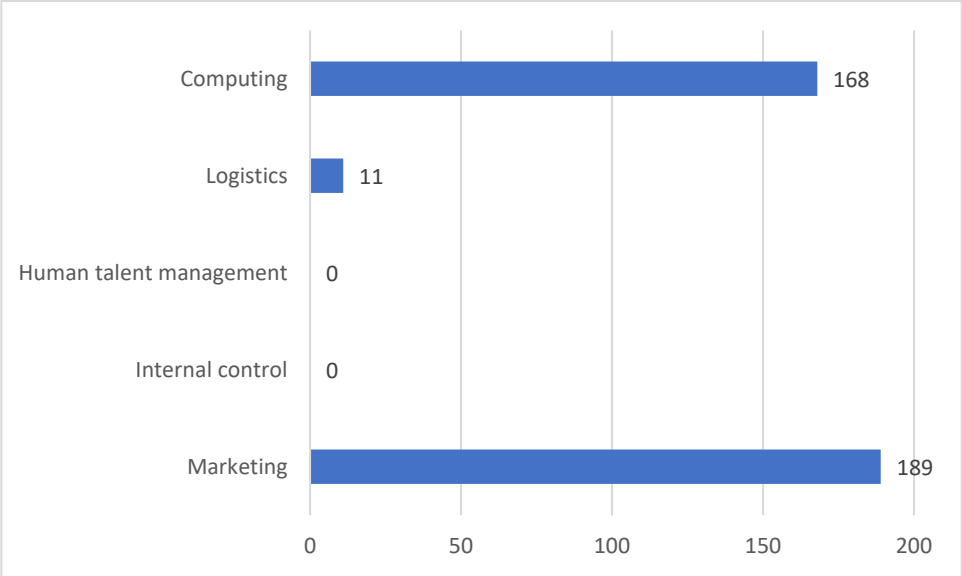

Figure 3. Business functions in which artificial intelligence is applied.

In these functional areas, professional software tools or mobile applications are often used; however, they are not considered forms of artificial intelligence, as they do not meet the defining criteria for AI systems. Examples of such applications include:

- Accounting automation
- Financial forecasting
- Fraud detection
- Candidate selection
- Payroll management
- Employee performance analysis
- Order and shipping automation
- Route optimization
- Cybersecurity
- Data protection
- Suspicious activity monitoring

Within the IT function, although a relatively higher frequency of AI usage is reported compared to other business areas, this usage is primarily concentrated in code assistant

tools designed for various software environments, such as Python, R, MATLAB, and Excel, among others.

Marketing is the area where the greatest variety of AI applications is observed, as illustrated in Figure 4. Most implementations are focused on advertising-related activities, with prominent uses including content generation, social media automation, and market analysis. Additional applications reported include customer data analysis and customer service.

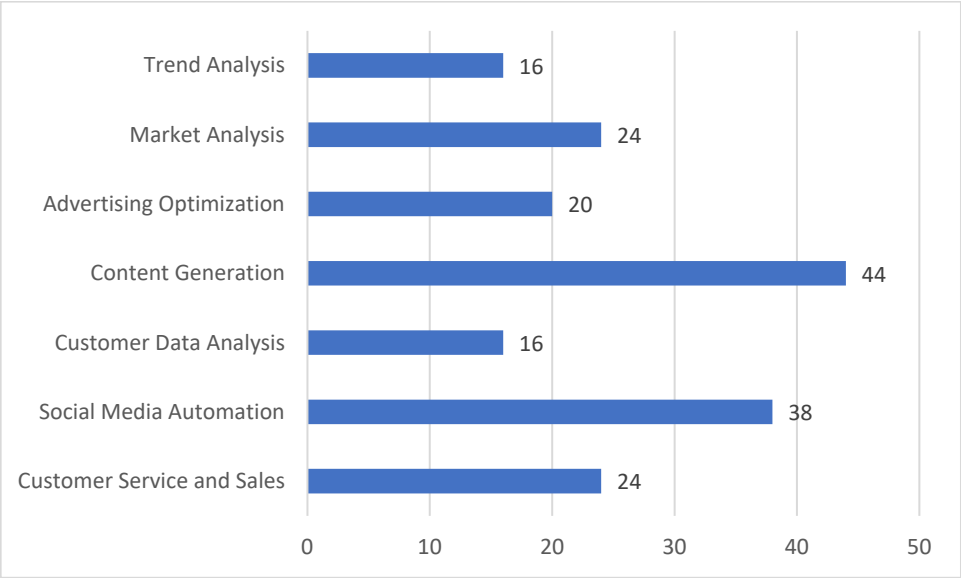

Figure 4. Use of artificial intelligence in marketing functions.

According to the data presented above, the specific AI tools used for each function were identified, as summarized in Table 4.

Table 4. AI tools used by business function.

Business Function	AI Tool(s) Used
Customer service and sales	ChatGPT
Social media automation	Hootsuite AI
Customer data analysis	HubSpot AI
Content generation	Copy.ai, Canva AI
Advertising optimization	Google Ads AI, Meta (Facebook) Ads AI
Market analysis	Brandwatch AI
Trend analysis	SEMrush AI
Inventory optimization	TradeGecko AI (QuickBooks Commerce)
Design and prototyping	Figma AI
Code assistants	ChatGPT

Some of the organizations under study reported using artificial intelligence tools for tasks such as: "...designing or improving advertising campaigns, generating code in programming languages such as R, Python, or MATLAB, and analyzing trends in campaign performance, among others".

Despite the aforementioned findings, it is important to note that, overall, the use of AI among the surveyed companies remains low.

An analysis of the barriers limiting AI adoption (see Figure 5) reveals that all identified obstacles were acknowledged by at least 27% of the enterprises, particularly those related to the availability of computing equipment. Among all reported barriers, the most frequently cited was the lack of professionals qualified to implement and manage AI technologies.

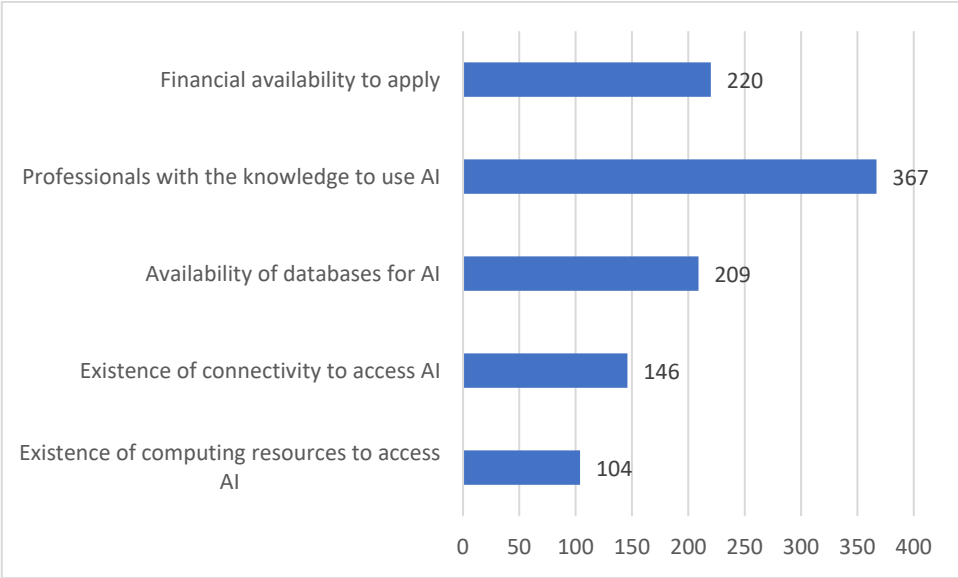

Figure 5. Frequency of recognition of barriers limiting AI adoption.

Table 5 presents the frequency of AI usage across different business functions. The most commonly used applications include social media automation, customer service, content generation for advertising, and market analysis. Less frequently used applications involve inventory optimization, design and prototyping, and code generation.

Table 5. Absolute and relative frequency of AI usage by sector.

AI Application Area	Services		Commerce		IT Sector		Industry	
	Count	%	Count	%	Count	%	Count	%
Customer service	12	7	7	6	3	7	1	4
Social media automation	18	11	11	12	4	9	1	4
Customer data analysis	9	5	5	4	2	4	0	0
Content generation	22	13	13	13	5	11	1	4
Advertising optimization	11	7	7	5	2	4	1	4
Market analysis	12	7	7	6	3	7	1	4
Trend analysis	9	5	5	4	2	4	0	0
Inventory optimization	0	0	0	0	0	0	1	4
Design and prototyping	0	0	0	0	5	11	2	9
Code assistants	0	0	0	0	17	39	13	57

Regarding inventory optimization and product design and prototyping, some companies reported that "...they have applied neural networks or genetic algorithms to optimize inventory levels and distribution routes...". These types of applications were mainly reported by organizations in the industrial and information technology sectors. The values in the table are presented in both absolute and relative frequencies, allowing for comparison across sectors.

In the evaluation of artificial intelligence adoption intensity, the observed behavior is summarized in Equation (3).

$$\text{IIIA} = \frac{\sum_{i=1}^{21} \text{Types of AI Utilization}}{21 * \text{n}} = \frac{367}{21 * 385} = 0.04539 \quad (3)$$

As observed, the values of the artificial intelligence adoption intensity indicator are low, particularly when considering the various ways in which AI technologies could be utilized or implemented within the organizations under study.

The logistic regression analysis between the AI adoption variable and the five identified barriers to implementation yielded acceptable results. Both the model coefficients and the pseudo-R^2 measures were high or within acceptable ranges (see Table 6). Furthermore, the Hosmer–Lemeshow test returned a significance value of $p = 0.213$ (>0.005), allowing the null hypothesis to be retained. This indicates that the model fits the data well, with no significant differences between observed and expected values.

Table 6. Model summary.

−2 Log Likelihood	Cox and Snell R Square		Nagelkerke R Square
58.980 *	0.707		0.945
Hosmer—Lemeshow Test			
Step	Chi-Square	df	Sig.
1	4.494	3	0.213

Note: * Estimation terminated at iteration number 7 because parameter estimates changed by less than 0.001.

Table 7 presents the behavior of the barriers that limit the frequency of AI use across different service sectors. As shown, the most frequently cited barrier is the lack of professionals with the necessary competencies to implement and operate AI technologies.

Table 7. Absolute and relative frequency of barriers limiting AI adoption by sector.

Barrier	Services		Commerce		IT Sector		Industry	
	Count	%	Count	%	Count	%	Count	%
Availability of computing resources for AI access	41	23	30	22	15	31	18	72
Availability of internet connectivity for AI	66	37	49	37	24	49	7	28
Availability of databases for AI	93	53	72	54	32	65	12	48
Professionals with knowledge to use AI	177	100	134	100	39	80	17	68
Financial resources to implement AI	98	55	76	22	33	67	13	57

Other highly influential barriers include the lack of available databases for analysis and insufficient financial resources to access AI services. Although the availability of computing equipment and internet connectivity were also recognized as barriers, their impact was comparatively lower than that of the other factors.

Additionally, AI usage was analyzed according to company size. Table 8 displays the absolute and relative frequency of AI adoption based on the size of the enterprise.

Table 8. Absolute and relative frequency of AI use by company size.

AI Application Area	Micro Enterprises		Small Enterprises		Medium Enterprises	
	Count	%	Count	%	Count	%
Customer service	20	5.48	3	18.75	1	25
Social media automation	30	8.22	7	43.75	1	25
Customer data analysis	14	3.84	2	12.5	0	0
Content generation	34	9.32	9	56.25	1	25
Advertising optimization	17	4.66	2	12.5	1	25
Market analysis	20	5.48	3	18.75	1	25
Trend analysis	14	3.84	2	12.5	0	0
Inventory optimization	0	0.00	0	0	1	25
Design and prototyping	3	0.82	1	6.25	1	25
Code assistants	25	6.85	3	18.75	2	50

In general, the data show that the percentage of AI utilization tends to increase with company size. Exceptions to this pattern include customer data analysis and trend analysis, which were not reported among medium-sized enterprises, possibly due to their lower representation in the overall sample. It is also noteworthy that content generation and social media automation show relatively higher usage rates compared to other AI functions analyzed.

Similarly, the behavior of the barriers to AI adoption was analyzed based on company size (see Table 9). In general, the data suggest that the recognition of barriers tends to increase with company size, particularly when comparing micro- and medium-sized enterprises. However, small enterprises generally show better results than micro enterprises in terms of overcoming these barriers.

Table 9. Absolute and relative frequency of barriers to AI adoption by company size.

Barrier	Micro Enterprises		Small Enterprises		Medium Enterprises	
	Count	%	Count	%	Count	%
Availability of computing resources for AI access	99	27.12	3	18.75	2	50
Availability of internet connectivity for AI	142	38.90	2	12.5	2	50
Availability of databases for AI	202	55.34	4	25	3	75
Professionals with knowledge to use AI	348	95.34	15	93.75	4	100
Financial resources to implement AI	212	58.08	5	31.25	3	75

The high levels of reported barriers among medium-sized companies may be attributed to their relatively low representation within the sample, even though their proportion reflects the overall population structure of the study.

In order to evaluate whether the observed differences in AI application levels and the barriers to adoption were statistically significant based on sector or company size, a one-way ANOVA test was conducted (see Table 10). According to the significance levels obtained, differences across sectors were statistically significant in only four application areas: social media automation, content generation, inventory optimization, and design and prototyping.

Table 10. One-way ANOVA: Differences in AI application by sector and company size.

AI Application/Barrier	Source of Variation	By Sector			By Company Size		
		df	F	Sig.	df	F	Sig.
Customer service	Between Groups	2	3.565	0.029	2	3.565	0.029
	Within Groups	382			382		
Social media automation	Between Groups	2	12.019	0.000	2	12.019	0.000
	Within Groups	382			382		
Customer data analysis	Between Groups	2	1.532	0.217	2	1.532	0.217
	Within Groups	382			382		
Content generation	Between Groups	2	18.557	0.000	2	18.557	0.000
	Within Groups	382			382		
Advertising optimization	Between Groups	2	2.581	0.077	2	2.581	0.077
	Within Groups	382			382		
Market analysis	Between Groups	2	3.565	0.029	2	3.565	0.029
	Within Groups	382			382		

Table 10. *Cont.*

AI Application/Barrier	Source of Variation	By Sector			By Company Size		
		df	F	Sig.	df	F	Sig.
Trend analysis	Between Groups	2	1.532	0.217	2	1.532	0.217
	Within Groups	382			382		
Inventory/order optimization	Between Groups	2	63.005	0.000	2	63.005	0.000
	Within Groups	382			382		
Design and prototyping	Between Groups	2	61.623	0.000	2	61.623	0.000
	Within Groups	382			382		
Code assistants	Between Groups	2	3.316	0.037	2	3.316	0.037
	Within Groups	382			382		
Availability of computers	Between Groups	2	.811	0.445	2	0.811	0.445
	Within Groups	382			382		
Internet connectivity	Between Groups	2	2.406	0.092	2	2.406	0.092
	Within Groups	382			382		
Database availability	Between Groups	2	3.221	0.041	2	3.221	0.041
	Within Groups	382			382		
Availability of professionals	Between Groups	2	0.142	0.868	2	0.142	0.868
	Within Groups	382			382		
Financial resources	Between Groups	2	2.530	0.081	2	2.530	0.081
	Within Groups	382			382		

Shaded cells are where significance is significant for having values below 0.05.

A similar pattern was observed when analyzing differences based on company size, with statistical significance identified in the same four application types. Additionally, a significant difference was found in the use of AI for customer service, which varied according to company size, suggesting that larger enterprises are more likely to have the resources required to implement such solutions.

Additionally, to verify the independence of samples based on sector and company size in relation to the barriers identified, a Kruskal–Wallis test was conducted. The results are summarized in Table 11.

Table 11. Kruskal–Wallis test results for independence of means by sector and company size.

Barrier	Significance by Sector	Significance by Company Size
Availability of computers	0.000	0.444
Internet connectivity	0.294	0.092
Availability of databases	0.388	0.041
Availability of qualified professionals	0.000	0.867
Financial resources	0.460	0.081

Shaded cells are where significance is significant for having values below 0.05.

As shown, the lack of computing equipment presents statistically significant differences by sector, an expected result considering that this barrier is likely to be less prevalent in the IT sector. Conversely, the absence of databases is only significantly differentiated

based on company size, which aligns with the understanding that larger enterprises tend to have more standardized processes and greater volumes of data.

Similarly, the availability of qualified professionals shows statistically significant differences across productive sectors, likely influenced by the higher presence of skilled personnel in the IT sector, where professionals are generally more prepared to implement AI technologies.

5. Discussion

This research provides a valuable empirical diagnosis, identifying specific applications of AI technologies, particularly those involving neural networks and large language models (LLMs), and offering insights into the barriers that currently hinder broader implementation.

In line with prior international studies, this research aligns with findings that describe the current state of AI adoption among SMEs globally [86]. However, unlike research that focuses on motivational drivers of adoption (e.g., Sharma et al. [87]), this study emphasizes structural and organizational barriers. These include the lack of qualified personnel, insufficient digital infrastructure (especially databases), and limited financial capacity—factors that were confirmed through logistic regression and supported by sector-based statistical analyses.

The results reaffirm that marketing is the most active functional area for AI use, particularly in social media automation [34], content generation, and market analysis. This supports the arguments presented by Li and Zhou [38] and Abrokwah-Larbi and Awuku-Larbi [93], who highlight the effectiveness of AI in marketing among SMEs. However, this pattern contrasts sharply with the almost complete absence of AI use in finance, logistics, and human resource management—areas that, according to studies such as Kumar et al. [96] and Syed et al. [97], have experienced considerable AI integration in other regions.

From a functional perspective, while tools like GitHub Copilot and Tabnine are gaining traction globally for accelerating software development [35], their use in Ecuadorian firms remains extremely limited. Even in the IT sector, adoption of these tools is mostly experimental. The data suggest a clear skills gap, with a strong correlation between AI adoption and the availability of technically trained professionals, mirroring findings from Priya et al. [40] in other emerging economies.

Company size also plays a role. The results indicate a moderate increase in AI usage among medium-sized enterprises, particularly in customer service and logistics. However, the differences are not linear. In fact, small enterprises in some cases show more favorable adoption behaviors than medium-sized ones, suggesting that resource availability alone does not determine adoption. Factors such as digital culture, strategic vision, and organizational maturity may mediate this relationship. This nuance supports literature indicating that technological absorption is not solely size-dependent, but contingent on broader systemic and strategic conditions [36].

Additionally, the analysis of statistically significant differences (ANOVA and Kruskal–Wallis tests) revealed that adoption patterns vary by sector and application type, particularly in areas such as content generation, inventory optimization, and design prototyping. These results emphasize the uneven maturity levels across business functions, with marketing functions showing clear early adoption, and others such as internal control and occupational safety exhibiting no AI usage at all.

Interestingly, while prior studies in comparable developing contexts (e.g., Dlamini and Schutte [8]) show higher AI adoption in accounting and financial domains, Ecuadorian MSMEs continue to rely on traditional tools that do not qualify as AI-based. This reflects not only a technological lag but also a conceptual misunderstanding of what constitutes AI, as many companies equate any digital system with AI.

The observed intensity of AI adoption, measured through a multidimensional indicator, was low, with a value of just 0.04. This confirms that not only the breadth but also the depth of AI implementation is limited.

This creates a significant opportunity for public and private stakeholders to intervene. From a policy perspective, differentiated strategies are needed: micro enterprises may benefit from basic digital literacy programs and awareness campaigns, while small- and medium-sized enterprises could be supported through financing instruments, data infrastructure development, and access to qualified human resources.

In addition, it is essential to consider the temporal dimension. The current landscape reflects an incipient phase of digital maturity, in which firms explore low-cost, low-complexity tools, primarily for marketing. Strategic planning for digital transformation, led by academia, government, and industry, can help transition these organizations into more advanced stages of AI integration across functions.

6. Limitations and Recommendations for Future Research

The exploratory and descriptive nature of this study limits its capacity to develop a comprehensive theoretical generalization. This limitation is reinforced by the low overall level of artificial intelligence (AI) adoption among Ecuadorian MSMEs, which constrains the possibility of conducting an in-depth comparative analysis.

The primary limitation of the study lies in the sample structure, particularly the relatively small representation of medium-sized enterprises. Although the sample was proportionally distributed, ensuring that the representation of medium-sized companies was consistent with both the general structure of the sample and their relative weight in the study population, it is important to acknowledge that this representation constitutes a limitation. This may affect the generalizability of certain findings, and future research should specifically address this issue.

Although AI adoption intensity was assessed, future research could benefit from evaluating this intensity using Likert-type scales that capture the frequency and sophistication of use, thereby providing a more granular understanding of AI maturity within organizations.

Future studies could further explore the behavioral and organizational variables influencing AI adoption decisions, as well as conduct longitudinal analyses to track changes over time. Mixed-methods research, including qualitative approaches such as case studies or in-depth interviews, would also be valuable to complement the quantitative findings and provide richer insights into the contextual factors shaping adoption pathways.

7. Conclusions

This study reaffirms the growing importance and expanding role of artificial intelligence (AI) across the various domains of business administration. However, in the specific context of Ecuadorian MSMEs, the level of AI adoption is still remarkably low and uneven, with clear limitations in both the scope of applications and the sectors involved. This result is supported by the value of the artificial intelligence adoption intensity indicator, which reached only 0.04. The highest levels of adoption were observed in marketing-related activities, particularly in content generation and social media automation.

While multiple barriers to AI implementation were identified, the most critical were the lack of qualified professionals capable of operating AI systems, and the absence of structured, accessible databases, which limits analytical depth and restricts broader application across business functions. Financial limitations were also recognized but appeared to have a comparatively lower impact.

The study also confirms that company size plays a role in AI adoption, with larger firms demonstrating a higher likelihood of integrating AI solutions due to greater resource availability. Nonetheless, this does not guarantee full adoption, as even within medium-sized enterprises, usage remains isolated and often exploratory.

In contrast to international findings—where AI has been widely adopted in financial management, internal control, and logistics—the Ecuadorian case reflects an early-stage scenario marked by limited digital maturity and sector-specific gaps, particularly in non-technical industries. This contrast underscores the need for strategic efforts from both public institutions and private initiatives to close these gaps.

From a managerial standpoint, the results highlight the urgent need to raise awareness of the practical benefits of AI and to promote its strategic implementation in business functions beyond marketing. The identification of barriers also provides actionable insights for designing intervention strategies, particularly in terms of workforce training and the development of data infrastructures, both of which are essential for unlocking the potential of AI in Ecuadorian MSMEs.

Author Contributions: Conceptualization, R.P.-C.; methodology, G.G.-V., M.D.M.-G. and R.P.-C.; validation, G.G.-V., R.P.-C. and M.D.M.-G.; formal analysis, A.S.-R. and R.P.-C.; investigation, G.G.-V., A.S.-R., R.P.-C., M.D.M.-G. and R.M.-V.; data curation, R.P.-C.; writing—original draft preparation, R.P.-C.; writing—review and editing, R.P.-C. and A.S.-R.; visualization, M.D.M.-G., A.S.-R. and R.P.-C.; supervision, R.M.-V.; project administration, G.G.-V. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: The study did not involve any clinical procedures, biomedical experimentation, or collection of sensitive personal data. Instead, the data were collected through anonymous surveys and interviews voluntarily completed by adult SME owner-managers, addressing only their business perceptions and general demographic characteristics. In Ecuador, according to Acuerdo Ministerial 4883 del Ministerio de Salud Pública (Registro Oficial Suplemento 173, del 12 de diciembre de 2013), ethical review by an Institutional Review Board (IRB) or Comité de Ética de Investigación en Seres Humanos (CEISH) is required only for biomedical or clinical research that may pose physical or psychological risks to participants. Our study, being observational, non-interventional, and of minimal risk, is exempt under this regulation. Nevertheless, we affirm that all procedures complied with the ethical standards of the 2013 revision of the Declaration of Helsinki, including respect for informed consent, privacy, and voluntary participation. Participants were informed of the purpose of the study and their right to withdraw at any point without consequence. No personal or identifiable information was recorded. The above is assumed to be an exemption from the ethical compliance requirement.

Informed Consent Statement: Verbal informed consent was obtained from all participants involved in the study. Prior to participation, respondents were informed about the purpose of the research, the voluntary nature of their participation, and the confidentiality of their responses. The study involved no sensitive personal data and was conducted in full compliance with the ethical principles outlined in the Declaration of Helsinki (2013 revision).

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Acknowledgments: The authors thank the anonymous reviewers of the journal for their extremely helpful suggestions to improve the quality of the article. The usual disclaimers apply.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Hajlaoui, Y.; Labib, R.; Plante, J.-F.; Gamache, M. Backpropagation-based inference for spatial interpolation to estimate the blastability index in an open pit mine. *Comput. Geosci.* **2025**, *194*, 105756. [CrossRef]
2. Redoloza, F.; Li, L.; Davis, A. Progressive growing generative adversarial networks using conditioning ratio for facies modeling in complex aquifers. *Hydrogeol. J.* **2023**, *31*, 1565–1580. [CrossRef]
3. Kim, S.H.; Schramm, S.; Adams, L.C.; Braren, R.; Bressem, K.K.; Keicher, M.; Platzek, P.-S.; Paprottka, K.J.; Zimmer, C.; Hedderich, D.M.; et al. Benchmarking the diagnostic performance of open source LLMs in 1933 Eurorad case reports. *Npj Digit. Med.* **2025**, *8*, 97. [CrossRef]
4. Xu, C.; Song, T.; Feng, W.; Li, X.; Ge, T.; Zheng, B.; Wang, L. *Accelerating Image Generation with Sub-Path Linear Approximation Model*; Lectures Notes in Computer Sciences; Leonardis, A., Ricci, E., Roth, S., Russakovsky, O., Sattler, T., Varol, G., Eds.; Springer: Berlin/Heidelberg, Germany, 2025; Volume 15111, pp. 323–339. [CrossRef]
5. Ma, R.; Cheng, Q.; Yao, J.; Peng, Z.; Yan, M.; Lu, J.; Liao, J.; Tian, L.; Shu, W.; Zhang, Y.; et al. Multimodal machine learning enables AI chatbot to diagnose ophthalmic diseases and provide high-quality medical responses. *Npj Digit. Med.* **2025**, *8*, 64. [CrossRef]
6. Giaccone, P.; D'Antoni, F.; Russo, F.; Ambrosio, L.; Papalia, G.F.; d'Angelis, O.; Vadalà, G.; Comelli, A.; Vollero, L.; Merone, M.; et al. Prevention and management of degenerative lumbar spine disorders through artificial intelligence-based decision support systems: A systematic review. *BMC Musculoskelet. Disord.* **2025**, *26*, 126. [CrossRef]
7. Pérez-Campdesuñer, R.; Sánchez-Rodríguez, A.; Garcia-Vidal, G.; Martínez-Vivar, R.; Ruiz-de la Peña, J.; De Miguel-Guzmán, M. The Premature Mortality of Micro and Small Enterprises. *Zagreb Int. Rev. Econ. Bus.* **2023**, *26*, 49–75. [CrossRef]
8. Dlamini, B.; Schutte, D.P. An Evaluation of the Adoption of Cloud Accounting by SMEs in Zimbabwe. *Int. J. Econ. Financ. Issues* **2025**, *15*, 288–294. [CrossRef]
9. Feijoó González, E.M.; Gutiérrez Jaramillo, N.D.; Medina Castillo, W.T.; Jaramillo Simbaña, R.M. Digital transformation in accounting for small and medium-sized enterprises in the province of El Oro, Ecuador. *Rev. Venez. Gerenc.* **2024**, *29*, 1580–1598. [CrossRef]
10. Jurado Jurado, J.C. Authority. Conceptual implications in management theories. *Cuad. Adm.* **2018**, *31*, 81–104. [CrossRef]
11. Felix, R. A Proposed Taxonomy of Management Systems. *Syst. Res. Behav. Sci.* **2003**, *20*, 21–29. [CrossRef]
12. Oña-Sinchiguano, B.E. Teoría de la Contingencia y Teoría de Costos en sectores económicos vulnerables. *Rev. Arbitr. Interdiscip. Koin.* **2020**, *5*, 706–716.
13. Paucar-Caceres, A.; Jerardino-Wiesenborn, B. A bridge for two views: Checkland's soft systems methodology and Maturana's ontology of the observer. *J. Oper. Res. Soc.* **2020**, *71*, 660–672. [CrossRef]
14. Hamid, S.R.; Isa, S.; Chew, B.C.; Altun, A. Quality Management Evolution from the Past to Present: Challenges for Tomorrow. *Organizacija* **2019**, *52*, 157–186. [CrossRef]
15. Thaichon, P.; Liyanaarachchi, G.; Quach, S.; Weaven, S.; Bu, Y. Online relationship marketing: Evolution and theoretical insights into online relationship marketing. *Mark. Intell. Plan.* **2019**, *38*, 676–698. [CrossRef]
16. Shen, W.; Hu, D.; Günay, E.E.; Okudan Kremer, G.E. Evolution of supply chain management: A sustainability focused review. *Int. J. Sustain. Manuf.* **2020**, *4*, 319–335. [CrossRef]
17. Obedgiu, V. Human resource management, historical perspectives, evolution and professional development. *J. Manag. Dev.* **2017**, *36*, 986–990. [CrossRef]
18. Reyes Espinosa, M.V.; de la Oliva de Con, F.; Sarduy González, M. Las relaciones financieras en la sociedad cubana actual. Experiencias y resultados [Financial relations in today's Cuban society. Experiences and results]. *An. Acad. Cienc. Cuba* **2022**, *12*, e1144. Available online: http://www.revistaccuba.cu/index.php/revacc/article/view/1144 (accessed on 11 May 2025).
19. Guerra Betancourt, K.; de Zayas Pérez, M.R.; González Guitián, M.V. Bibliometric analysis of publications related to innovation projects and their management in Scopus, 2001–2011. *Rev. Cuba. Inf. Cienc. Salud* **2013**, *24*, 281–294.
20. Ormazabal, M.; Rich, E.; Sarriegi, J.M.; Viles, E. Environmental Management Evolution Framework: Maturity Stages and Causal Loops. *Organ. Environ.* **2017**, *30*, 27–50. [CrossRef]
21. Benzaquen, J.; Charles, V. A stratified bootstrapping approach to assessing the success of TQM implementation in Peruvian companies. *Total Qual. Manag. Bus. Excell.* **2020**, *33*, 178–201. [CrossRef]
22. Shubik, M. Bibliography on Simulation, Gaming, Artificial Intelligence and Allied Topics. *J. Am. Stat. Assoc.* **1960**, *55*, 736–751. [CrossRef]
23. Sparrow, R. Friendly AI will still be our master. Or, why we should not want to be the pets of super-intelligent computers. *AI Soc.* **2024**, *39*, 2439–2444. [CrossRef]
24. Gao, J.; Fan, C.; Xu, L.; Chen, H.; Chen, H.; Liang, Z. Intelligent decision making and risk management in stock index futures markets under the influence of global geopolitical volatility. *Omega* **2025**, *133*, 103272. [CrossRef]
25. Soundar Rajan, P.; Arul Prasanna, M.; Nishar Ahamed, M.; Sathishkumar, R. Enhancing the performance and economic efficiency of range-extended electric vehicles: A hybrid dual stream spectrum deconvolution neural network with Beluga Whale Optimization. *J. Energy Storage* **2024**, *104*, 114450. [CrossRef]

26. Sanju, P.; Ahmed, N.S.S.; Ramachandran, P.; Sajid, P.M.; Jayanthi, R. Enhancing thyroid disease prediction and comorbidity management through advanced machine learning frameworks. *Clin. eHealth* **2025**, *8*, 7–16. [CrossRef]
27. Malaikrisanachalee, S.; Wongwai, N.; Butsan, N. Optimal Location for Tower Crane and Dynamic Trailer Parking in High-Rise Modular Building Construction Project. *Int. J. Math. Eng. Manag. Sci.* **2024**, *9*, 965–987. [CrossRef]
28. Fahland, D.; Fournier, F.; Limonad, L.; Skarbovsky, I.; Swevels, A.J.E. How well can a large language model explain business processes as perceived by users? *Data Knowl. Eng.* **2025**, *157*, 102416. [CrossRef]
29. Du, K.; Zhao, Y.; Mao, R.; Xing, F.; Cambria, E. Natural language processing in finance: A survey. *Inf. Fusion* **2025**, *115*, 102755. [CrossRef]
30. Tomczyk, P.; Brüggemann, P.; Vrontis, D. AI meets academia: Transforming systematic literature reviews. *EuroMed J. Bus.* 2024, *ahead of print*. [CrossRef]
31. Khan, N.; Khan, W.; Humayun, M.; Naz, A. (Eds.) Unlocking the potential: Artificial intelligence applications in sustainable tourism. In *The Role of Artificial Intelligence in Regenerative Tourism and Green Destinations*; Emerald Group Publishing Ltd.: Leeds, UK, 2024; pp. 303–316. [CrossRef]
32. Ngo, T.-L.; Pham, K.-L.; Cao, M.-S.; Pham, S.-B.; Phan, X.-H. Dialogue Act Segmentation for Vietnamese Human-Human Conversational Texts. In Proceedings of the 9th International Conference on Knowledge and Systems Engineering (KSE), Hue, Vietnam, 19–21 October 2017; pp. 203–208. [CrossRef]
33. Praveen Kumar, T.; Reddy, R.V.T.; Ariwa, E. Customer-Centric AI Solutions. In *Minds Unveiled: Exploring the Effects of Generative AI on Business Behavior*; Rodriguez, R.V., Hemachandran, K., Eds.; Taylor and Francis: Oxfordshire, UK, 2025; pp. 58–69. [CrossRef]
34. Yen, J.C.; Wang, T. A Topic Modeling-Based Review of Digital Transformation Literature in Accounting. In *Digital Transformation in Accounting and Auditing: Navigating Technological Advances for the Future*; Perdana, A., Wang, T., Eds.; Palgrave Macmillan: London, UK, 2024; pp. 105–118. [CrossRef]
35. Corso, V.; Mariani, L.; Micucci, D.; Riganelli, O. Assessing AI-Based Code Assistants in Method Generation Tasks. In Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering, Lisbon, Portugal, 14–20 April 2024; pp. 380–381. [CrossRef]
36. Pei, X.; Pei, J.; Hou, H.; Zhao, Y. Optimizing casting process using a combination of small data machine learning and phase-field simulations. *Npj Comput. Mater.* **2025**, *11*, 27. [CrossRef]
37. Chong, L.; Lo, I.-P.; Rayan, J.; Dow, S.; Ahmed, F.; Lykourentzou, I. Prompting for products: Investigating design space exploration strategies for text-to-image generative models. *Des. Sci.* **2025**, *11*, e2. [CrossRef]
38. Li, Q.; Zhou, E. Design and Implementation of Automatic Generation Algorithm for Advertising Artistic Design Based on Neural Networks. *Comput.-Aided Des. Appl.* **2024**, *21*, 114–127. [CrossRef]
39. Pishgar, M.; Issa, S.F.; Sietsema, M.; Pratap, P.; Darabi, H. Redeca: A novel framework to review artificial intelligence and its applications in occupational safety and health. *Int. J. Environ. Res. Public Health* **2021**, *18*, 6705. [CrossRef] [PubMed]
40. Priya, P.K.; Rudra, K.; Kandula, D.S.; Teja, G.C.; Reddy, R.V.T. 13 An innovative analysis of AI-powered automation techniques for business management. In *Toward Artificial General Intelligence: Deep Learning, Neural Networks, Generative AI*; de Albuquerque, V., Raj, P., Yadav, S., Eds.; De Gruyter: Berlin, Germany, 2024; pp. 269–286. [CrossRef]
41. Steinmacher, I.; Penney, J.M.; Felizardo, K.R.; Garcia, A.F.; Gerosa, M.A. Can ChatGPT emulate humans in software engineering surveys? In Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement, Bacelona, Spain, 24–25 October 2024; pp. 414–419. [CrossRef]
42. Musa, A.M.H.; Lefkir, H. The role of artificial intelligence in achieving auditing quality for small and medium enterprises in the Kingdom of Saudi Arabia. *Int. J. Data Netw. Sci.* **2024**, *8*, 835–844. [CrossRef]
43. Terstegen, S.; Schmalen, B.; Hinz, A.; Pricelius, M. Introductory Qualification on Artificial Intelligence: Productive and Humane Work Design with AI in Small and Medium-Sized Enterprises. In *Work and AI 2030: Challenges and Strategies for Tomorrow's Work*; Knappertsbusch, I., Gondlach, K., Eds.; Springer: Berlin/Heidelberg, Germany, 2023; pp. 343–351. [CrossRef]
44. Hommerson, S.; Gallese, C.; Garel, F.; Aleksovska, A.; Bikas, H. Ethics, Data and Privacy Management for AI Solutions in Manufacturing at SMEs (AIRISE). In *Mechanical Engineering*; Lecture Notes; Alexopoulos, K., Makris, S., Stavropoulos, P., Eds.; Springer: Berlin/Heidelberg, Germany, 2025; pp. 279–286. [CrossRef]
45. Sargut, D.K. Study on the effects of digitisation in small and medium-sized German companies. *Calitatea* **2019**, *20*, 561–566.
46. Soomro, R.B.; Memon, S.G.; Dahri, N.A.; Al-Rahmi, W.M.; Aldriwish, K.A.; Salameh, A.; Al-Adwan, A.S.; Saleem, A. The Adoption of Digital Technologies by Small and Medium-Sized Enterprises for Sustainability and Value Creation in Pakistan: The Application of a Two-Staged Hybrid SEM-ANN Approach. *Sustainability* **2024**, *16*, 7351. [CrossRef]
47. Wolf-Brenner, C.; Pammer-Schindler, V.; Breitfuss, G. How Do Professionals in SMEs Engage with AI and Regulation? An Interview Study in Austria. In Proceedings of the Mensch und Computer 2024, Karlsruhe, Germany, 1–4 September 2024; Association for Computing Machinery: New York, NY, USA, 2024; pp. 646–650. [CrossRef]

48. Mokwena, S.M.; van der Poll, J.A. The Impact of the Fourth Industrial Revolution (4IR) on Small and Medium Enterprises (SMEs) in South Africa's Tourism Sector. In *Business Sustainability with Artificial Intelligence (AI): Challenges and Opportunities. Studies in Systems, Decision and Control*; AlDhaen, E., Braganza, A., Hamdan, A., Chen, W., Eds.; Springer: Cham, Switzerland, 2025; Volume 568, pp. 239–249. [CrossRef]
49. Pozzo, D.N.; Gonzalez Beleno, C.A.; Correa, K.R.; Donado, M.G.; Gomez Pedroza, F.J.; Moncada Diaz, J.E. Managers' attitudes and behavioral intentions towards using artificial intelligence for organizational decision-making: A study with Colombian SMEs. *Procedia Comput. Sci.* **2024**, *238*, 956–961. [CrossRef]
50. Schwaeke, J.; Peters, A.; Kanbach, D.K.; Kraus, S.; Jones, P. The new normal: The status quo of AI adoption in SMEs. *J. Small Bus. Manag.* **2024**, *63*, 1297–1331. [CrossRef]
51. Hüsson, D.; Holland, A.; Fathi, M.; Arteaga Sanchez, R. Analysis and illustration of the practical impact of Artificial Intelligence and Intelligent Personal Assistants on business processes in small- and medium-sized service enterprises. In Proceedings of the 2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC), Melbourne, Australia, 17–21 October 2021; pp. 3303–3310. [CrossRef]
52. Lada, S.; Chekima, B.; Karim, M.R.A.; Fabeil, N.F.; Ayub, M.S.; Amirul, S.M.; Ansar, R.; Bouteraa, M.; Fook, L.M.; Zaki, H.O. Determining factors related to artificial intelligence (AI) adoption among Malaysia's small and medium-sized businesses. *J. Open Innov. Technol. Mark. Complex.* **2023**, *9*, 100144. [CrossRef]
53. Okuno, M.J.; Okuno, H.G. Legal frameworks for AI service business participants: A comparative analysis of liability protection across jurisdictions. *AI Soc.* **2025**. [CrossRef]
54. Yuan, L.Y.; Alias, N.; Yaacob, M.H. What factors influence SMEs' equity financing: A systematic literature review and future agendas. *Multidiscip. Rev.* **2025**, *8*, 2025199. [CrossRef]
55. Hermansyah, Y. Assessing the Impact of Communicative Artificial Intelligence Based Accounting Information Systems on Small and Medium Enterprises. *J. Wirel. Mob. Netw. Ubiquitous Comput. Dependable Appl.* **2023**, *14*, 230–239. [CrossRef]
56. Rasmus, K. Artificial Intelligence Working to Secure Small Enterprises. In *Artificial Intelligence for Security: Enhancing Protection in a Changing World*; Sipola, T., Alatalo, J., Wolfmayr, M., Kokkonen, T., Eds.; Springer Nature: Berlin/Heidelberg, Germany, 2024; pp. 165–188. [CrossRef]
57. Pu, L.; Radics, R.I.; Umar, M.; Quan, Z. Investigating How Chinese E-Commerce SMEs Would Adopt AI Live-Streaming via a Dual Analysis: Trendy and Transformative. *J. Organ. End User Comput.* **2024**, *36*, 1–22. [CrossRef]
58. Tiwari, P.A. Moderating Role of Supply Chain Re-engineering Capabilities on Artificial Intelligence-Based Risk Management and Supply Chain Agility: A Resource-Based View. In *Anticipating Future Business Trends: Navigating Artificial Intelligence Innovations. Studies in Systems, Decision and Control*; El Khoury, R., Ed.; Springer: Cham, Switzerland, 2024; Volume 535. [CrossRef]
59. Khan, M.S.; Abdullah, O.; Verma, A.K.; Designation, S. AI and Machine Learning Applications in Tourism: An Empirical Model-Based Research. In Proceedings of the 4th International Conference on Technological Advancements in Computational Sciences (ICTACS), Tashkent, Uzbekistan, 13–15 November 2024; pp. 1167–1172. [CrossRef]
60. Saleem, I.; Al-Breiki, N.S.S.; Asad, M. The nexus of artificial intelligence, frugal innovation and business model innovation to nurture internationalization: A survey of SME's readiness. *J. Open Innov. Technol. Mark. Complex.* **2024**, *10*, 100326. [CrossRef]
61. Brückner, A.; Hein, P.; Hein-Pensel, F.; Mayan, J.; Wölke, M. Human-Centered HCI Practices Leading the Path to Industry 5.0: A Systematic Literature Review. In *Communications in Computer and Information Science*; Stephanidis, C., Antona, M., Ntoa, S., Salvendy, G., Eds.; Springer: Berlin/Heidelberg, Germany, 2023; Volume 1832, pp. 3–15. [CrossRef]
62. Bendimerad, A.; Remil, Y.; Mathonat, R.; Kaytoue, M. On-Premise AIOps Infrastructure for a Software Editor SME: An Experience Report. In Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, San Francisco, CA, USA, 3–9 December 2023; pp. 1820–1831. [CrossRef]
63. Rawindaran, N.; Jayal, A.; Prakash, E. Machine learning cybersecurity adoption in small and medium enterprises in developed countries. *Computers* **2021**, *10*, 150. [CrossRef]
64. Dangi, D.; Bhagat, A.; Dixit, D.K. Sentiment analysis of social media data based on chaotic coyote optimization algorithm based time weight-AdaBoost support vector machine approach. *Concurr. Comput. Pract. Exp.* **2022**, *34*, e6581. [CrossRef]
65. Santana, L.E.; Cánepa, G.H. ¿Son bots? Automatización en redes sociales durante las elecciones presidenciales de Chile 2017. *Cuadernos. Info.* **2019**, *44*, 61–77. [CrossRef]
66. Thapa, S.; Shiwakoti, S.; Shah, S.B.; Adhikari, S.; Veeramani, H.; Nasim, M.; Naseem, U. Large language models (LLM) in computational social science: Prospects, current state, and challenges. *Soc. Netw. Anal. Min.* **2025**, *15*, 4. [CrossRef]
67. Haji, R.; Stock, W.G. User settings for advertising optimization on Facebook: Active customer participation or settings blindness? *Telemat. Inform.* **2021**, *59*, 101548. [CrossRef]
68. Wang, K. Multifactor prediction model for stock market analysis based on deep learning techniques. *Sci. Rep.* **2025**, *15*, 5121. [CrossRef]
69. Wang, Y.J.; Choo, W.C.; Ng, K.Y.; Bi, R.; Wang, P.W. Evolution of AI enabled healthcare systems using textual data with a pretrained BERT deep learning model. *Sci. Rep.* **2025**, *15*, 7540. [CrossRef]

70. Artene, A.E.; Domil, A.E. Neural Networks in Accounting: Bridging Financial Forecasting and Decision Support Systems. *Electronics* **2025**, *14*, 993. [CrossRef]
71. Fu, P.; Yang, H.; Qian, W.; Mohamed, E.I.; Almohri, W.A.J.; Alshanbari, H.M. Financial engineering and the digital economy: The implementations of machine learning algorithms. *Alex. Eng. J.* **2025**, *125*, 311–319. [CrossRef]
72. Hafez, I.Y.; Hafez, A.Y.; Saleh, A.; Abd El-Mageed, A.A.; Abohany, A.A. A systematic review of AI-enhanced techniques in credit card fraud detection. *J. Big Data* **2025**, *12*, 6. [CrossRef]
73. Trovão, H.; Mamede, H.S.; Trigo, P.; Santos, V. SMEs Recruitment Processes Supported by Artificial Intelligence: A Position Paper. In *Lecture Notes in Networks and System*; Iglesias, A., Shin, J., Patel, B., Joshi, A., Eds.; Springer: Berlin/Heidelberg, Germany, 2024; Volume 834, pp. 179–191. [CrossRef]
74. Susitra, D.; Pratheesh, G.; Karthiga, J.D.; Dhanalakshmi, K.; Shamreen Ahamed, B.; Sylvia Grace, J. Implementing HR Management System with an AI Powered ChatBot. In Proceedings of the 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things, IDCIoT, Bengaluru, India, 5–7 February 2025; pp. 2166–2172. [CrossRef]
75. Ravisankar, A.; Kulkarni, A.; Takore, T.T.; Hasan, D.S.; Dharani, K.; Naganjaneyulu, K.V. Using natural language processing and the internet of things for employee performance evaluation. In *Interdisciplinary Approaches to AI, Internet of Everything, and Machine Learning*; Pandey, D., Muniandi, B., Pandey, B., George, A., Eds.; IGI Global: Hershey, PA, USA, 2025; pp. 383–395. [CrossRef]
76. Tsintotas, K.A.; Moutsis, S.N.; Kansizoglou, I.; Konstantinidis, F.K.; Gasteratos, A. The Advent of AI in Modern Supply Chain. In *Communications in Computer and Information Science*; Kostavelis, I., Folinas, D., Aidonis, D., Achillas, C., Eds.; Springer: Berlin/Heidelberg, Germany, 2025; Volume 2110, pp. 333–343. [CrossRef]
77. Patnaik, S.; Patnaik, A. Advancements in production planning and control. In *Automation in Garment Manufacturing*; Nayak, R., Padhye, R., Eds.; Elsevier: Amsterdam, The Netherlands, 2018; pp. 291–309. [CrossRef]
78. Chen, F.; Zhu, Q. Intelligent optimization method for hazardous materials transportation routing with multi-mode and multi-criterion collaborative constraints. *Sci. Rep.* **2025**, *15*, 7804. [CrossRef]
79. Tyagi, A.K.; Tiwari, S.; Arumugam, S.K.; Sharmaç, A.K. *Artificial Intelligence-Enabled Digital Twin for Smart Manufacturing*; Wiley & Sons: Hoboken, NJ, USA, 2025. [CrossRef]
80. Li, Y.; Peng, Y.; Huo, Y.; Lyu, M.R. Enhancing LLM-Based Coding Tools through Native Integration of IDE-Derived Static Context. In Proceedings of the 2024 International Workshop on Large Language Models for Code, LLM4Code, Lisbon, Portugal, 20 April 2024; pp. 70–74. [CrossRef]
81. Zhang, J.; Bu, H.; Wen, H.; Liu, Y.; Fei, H.; Xi, R.; Li, L.; Yang, Y.; Zhu, H.; Meng, D. When LLMs meet cybersecurity: A systematic literature review. *Cybersecurity* **2025**, *8*, 55. [CrossRef]
82. Almotiri, S.H. AI driven IOMT security framework for advanced malware and ransomware detection in SDN. *J. Cloud Comput.* **2025**, *14*, 19. [CrossRef]
83. Raja, D.P.; Sheelambigai, P.; Sruthi Meera, P.; Valarmathi, K. Suspicious Activity Monitoring System using Machine Learning. In Proceedings of the 8th International Conference on Communication and Electronics Systems, ICCES 2023, Coimbatore, India, 1–3 June 2023. [CrossRef]
84. Yüksel, S.; Eti, S.; Dinçer, H.; Gökalp, Y.; Olaru, G.O.; Oflaz, N.K. Innovative financial solutions for sustainable investments using artificial intelligence-based hybrid fuzzy decision-making approach in carbon capture technologies. *Financ. Innov.* **2025**, *11*, 20. [CrossRef]
85. Baabdullah, A.; Alalwan, A.; Slade, E.; Raman, R.; Khatatneh, K. SMEs and Artificial Intelligence (AI): Antecedents and Consequences of AI-based B2B Practices. *Ind. Mark. Manag.* **2021**, *98*, 255–270. [CrossRef]
86. Ayinaddis, S.G. Artificial intelligence adoption dynamics and knowledge in SMEs and large firms: A systematic review and bibliometric analysis. *J. Innov. Knowl.* **2025**, *10*, 100682. [CrossRef]
87. Sharma, S.; Singh, G.; Islam, N.; Dhir, A. Why Do SMEs Adopt Artificial Intelligence-Based Chatbots? *IEEE Trans. Eng. Manag.* **2024**, *71*, 1773–1786. [CrossRef]
88. Arranz, C.F.A.; Arroyabe, M.F.; Arranz, N.; Fernandez de Arroyabe, J.C. Digitalisation dynamics in SMEs: An approach from systems dynamics and artificial intelligence. *Technol. Forecast. Soc. Change* **2023**, *196*, 122880. [CrossRef]
89. Wei, R.; Pardo, C. Artificial intelligence and SMEs: How can B2B SMEs leverage AI platforms to integrate AI technologies? *Ind. Mark. Manag.* **2022**, *107*, 466–483. [CrossRef]
90. Carayannis, E.G.; Dumitrescu, R.; Falkowski, T.; Papamichail, G.; Zota, N.R. Enhancing SME resilience through artificial intelligence and strategic foresight: A framework for sustainable competitiveness. *Technol. Soc.* **2025**, *81*, 102835. [CrossRef]
91. Khan, A.N.; Mehmood, K.; Soomro, M.A. Knowledge Management-Based Artificial Intelligence (AI) Adoption in Construction SMEs: The Moderating Role of Knowledge Integration. *IEEE Trans. Eng. Manag.* **2024**, *71*, 10874–10884. [CrossRef]
92. Al Dhaheri, M.H.; Ahmad, S.Z.; Papastathopoulos, A. Do environmental turbulence, dynamic capabilities, and artificial intelligence force SMEs to be innovative? *J. Innov. Knowl.* **2024**, *9*, 100528. [CrossRef]
93. Abrokwah-Larbi, K.; Awuku-Larbi, Y. The impact of artificial intelligence in marketing on the performance of business organizations: Evidence from SMEs in an emerging economy. *J. Entrep. Emerg. Econ.* **2024**, *16*, 1090–1117. [CrossRef]

94. Widiawati, H.S.; Ramadhani, R.A.; Sumantri, B.A. The role of eco-digital learning in enhancing the impact of IoT, blockchain, and artificial intelligence on green supply chain for SME internationalization. *Probl. Perspect. Manag.* **2025**, *23*, 76–89. [CrossRef]
95. Cubric, M.; Li, F. Bridging the 'Concept–Product' gap in new product development: Emerging insights from the application of artificial intelligence in FinTech SMEs. *Technovation* **2024**, *134*, 103017. [CrossRef]
96. Kumar, M.; Raut, R.D.; Mangla, S.K.; Ferraris, A.; Choubey, V.K. The adoption of artificial intelligence powered workforce management for effective revenue growth of micro, small, and medium scale enterprises (MSMEs). *Prod. Plan. Control* **2022**, *35*, 1639–1655. [CrossRef]
97. Syed, M.H.; Khan, S.; Rabbani, M.R.; Thalassinos, Y.E. An artificial intelligence and NLP based Islamic FinTech model combining zakat and Qardh-Al-Hasan for countering the adverse impact of COVID 19 on SMEs and individuals. *Int. J. Econ. Bus. Adm.* **2020**, *8*, 351–364. [CrossRef]
98. Wang, J.; Lu, Y.; Fan, S.; Hu, P.; Wang, B. How to survive in the age of artificial intelligence? Exploring the intelligent transformations of SMEs in central China. *Int. J. Emerg. Mark.* **2021**, *17*, 1143–1162. [CrossRef]
99. Chatterjee, S.; Chaudhuri, R.; Vrontis, D.; Thrassou, A.; Ghosh, S.K. Adoption of artificial intelligence-integrated CRM systems in agile organizations in India. *Technol. Forecast. Soc. Change* **2021**, *168*, 120783. [CrossRef]
100. Ferreira dos Santos, J.P.; de Matos, C.A.; Groznik, A. The role of artificial intelligence in smart city systems usage: Drivers, barriers, and behavioural outcomes. *Technol. Soc.* **2025**, *81*, 102867. [CrossRef]

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